# Software Testing Plan

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CSE General

## 1. Introduction

This document outlines the overall testing strategy, scope, and objectives for the E-Book Management System.
The goal of testing is to ensure that the system functions correctly, meets user requirements, and is free from critical defects before deployment.

## 2. Test Strategy

The following testing levels are used in the project:

- **Unit Testing:**
  Individual modules such as login, book upload, search, and user management are tested independently.
- **Integration Testing:**
  Interaction between modules (frontend, backend, and database) is tested to ensure smooth data flow.
- **System Testing:**
  The complete E-Book Management System is tested as a whole to verify functional and non-functional requirements.
- **Acceptance Testing:**
  Final testing is performed to confirm that the system meets user and project requirements.

## 3. Test Environment

The test environment specifies the resources required for testing.

- **Hardware:** Personal computer or laptop
- **Software:** Web browser, database server, application server
- **Test Data:** Sample user accounts, book records, and search queries

## 4. Test Cases

Test cases are designed to validate system functionality.

Each test case includes:

- Input data
- Expected output
- Actual output
- Pass/Fail criteria

This ensures systematic and repeatable testing.

## 5. Defect Management

Defect Management defines the process for handling errors found during testing.

- Defects are logged with details such as description and severity
- Developers analyse and fix reported defects
- Retesting is performed after fixes
- Ensures improved software quality and reliability